PILOTING AND VALIDATING THE CLIMATE SECURITY SENSITIVITY SCORING TOOL (CSST)

Evidence from the field on the climate security sensitivity of the Climate Smart Village approach in Cinzana, Mali

Validation Report

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Titles in this series aim to disseminate interim research on the scaling of climate services and climate-smart agriculture in Africa, in order to stimulate feedback from the scientific community.

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Cover photo: © CIAT/Carolina Sarzana – Ngakoro Climate Smart Village farmers, Mali

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ABSTRACT

A one-day workshop in Ségou, Mali was organized to test the results of the Climate Security Sensitivity Tool (CSST) and reflect on the reliability of its recommendations. This document reports on the results of the CSST piloted on the Climate Smart Village (CSV) approach implemented in Cinzana and on the outcomes of the workshop that reflected upon these results and recommendations. The workshop validated most drivers of conflict and insecurity flagged by the CSST, while pointing out that the human hazards risk indicator was overestimated since most conflict experienced are resource-related rather than linked to the threats of armed groups as well as the fact that the vulnerable groups indicator does not fully capture the complexity of marginalized groups in the region. The workshop highlighted the program's achievements, including advancements in agricultural technologies, gender empowerment through market gardening, and improved community collaboration. However, shortcomings in addressing institutional natural resource management and awareness-raising initiatives were noted. Participants emphasized the repercussions, such as deforestation and resource conflicts, signaling the need to prioritize institution-building and education in CSV programs. Aligning with CSST recommendations, the findings underscore the significance of integrating peace-focused elements to bolster climate adaptation strategies and prevent resource-related conflicts in vulnerable contexts. The workshop's outcomes validated the CSST's utility and demonstrated that the tool can provide suitable recommendations, especially when coupled with localized insights, enhancing its value in crafting context-specific climate adaptation programs that effectively address the drivers of conflict and insecurity in vulnerable regions.

Keywords

Climate security; climate-smart agriculture; climate change; peacebuilding; agriculture; adaptation.

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1. INTRODUCTION

Climate adaptation interventions, such as programs promoting climate-smart agricultural innovations, are proving effective in increasing farmer resilience as well as food and nutrition security (Mizik, 2021; Thornton et al., 2022). However, there is often little understanding of the potential positive and negative externalities that these programs can have, particularly in terms of peace and security. Maladaptation is the process whereby improperly built adaptation strategies can result in more vulnerability of other systems, sectors or social groups (Schipper, 2020; Barnett & O'Neill, 2010). It can create and sustain lock-ins, magnify inequity, marginalize people, and places vulnerable to climate-related risks, such as low-income households, people who reside in informal settlements, ethnic minorities, Indigenous Peoples among others (IPCC, 2022). These are commonly recognized drivers of conflict which must be accounted for while designing programs to avoid creating or exacerbating conflicts. Acknowledging the interlinkages between climate action, natural resource use and peace and security is fundamental for integrating climate and conflict sensitive programming interventions. Maladaptive climate initiatives neglecting those associations can foster power asymmetries, grievances, and competition for resources, especially in conflict-affected and fragile contexts (Moran et al., 2018; Krampe et al., 2021).

The CGIAR Focus Climate Security has been contributing to bridge this gap through the development of a Climate Security Sensitivity Tool (CSST), a programming assessment tool for conflict-sensitive and peace-responsive climate action in agricultural interventions. The CSST is a means for change agents from governmental and non-governmental organizations to support rural communities to adapt to climate change while reducing the potential for conflict of their programs and maximize social cohesion and integration. It is useful for stakeholders investing in and designing an agricultural climate action program with the goal of preventing maladaptation and unintended consequences.

The CSST is meant to be adopted at the designing phase of a project. It aims to improve the suitability of agricultural climate adaptation program designs in relation to these pre-existing drivers of conflict and insecurity, and to make recommendations on how they can be more effectively implemented. It does so by prioritizing climate-peace mechanisms based on contextual drivers of conflict and insecurity.

A workshop was held in Ségou, in Mali on the 14th of July 2023, which reunited experts on different social, political, economic, and agricultural characteristics of this region. The goal of this workshop was to test the results of the CSST and reflect on the reliability of its recommendations. This document therefore aims to report on (1) the results of the CSST piloted on a climate adaptation program, the Climate Smart Village (CSV) approach implemented in Cinzana, within the Ségou district, and on (2) the outcomes of the workshop that reflected upon these results and recommendations.

Firstly, this report provides the theoretical background and methodology underlying the CSST, it then provides introductory information on the case study through describing the characteristics of Cinzana's municipality, and the Ségou region more broadly, and of the CSV approach thereby implemented. It deploys the CSST on this case study providing recommendations for improving the conflict-sensitiveness and peace responsiveness of this climate action program. Lastly, it reports on the CSST validation workshop held in Ségou, which gathered key informants in the region to discuss and fine-tune the CSST results.

2. CSST THEORETICAL APPROACH

The CSST is employed on the premise that any fragile context is characterized by a unique set of risk factors for crises that can lead to insecurity and conflict, including natural hazards, human hazards, socioeconomic vulnerabilities, vulnerable groups, low institutional capacity to cope with shocks, and infrastructural coping capacity. These risk factors are retrieved from the crisis risk model developed by the Joint Research Center (JRC) of the European Commission, the INFORM Risk model (Marin-Ferrer et al., 2017).

Crisis risk models are particularly relevant for characterizing the contextual drivers of conflict and insecurity of different geographies as they assess threats to humans' wellbeing from a systemic approach while considering the complexity of the interlinkages between natural and socio-political vulnerability factors, leading to fragility and in some cases to conflict (Borodzicz, 2005; Simpson et al., 2021; Stein & Walch, 2017). Socioeconomic vulnerabilities inform on the extent to which poverty and inequalities are pressuring in a given locality, therefore capturing grievance-related drivers of conflict (Kett & Rowson, 2007). The vulnerable groups risk category appraises levels of domestic food price levels and volatility, food insecurity, the prevalence of undernourishment and the number of refugees and displaced people, being factors indicating levels of exclusion and marginalization, as well as insurrection, which all can lead to insecurity (Stein & Walch, 2017; Walch, 2018; Bora et al., 2011; Thalheimer & Webersik, 2020). The model's risk category on the lack of institutional coping capacity is composed of indicators defining governments' effectiveness and corruption as well as their capacity to cope with disasters and shocks. Poor institutional capacity to cope with shocks is widely recognized as a conflict and insecurity driver, as institutional and political landscapes determine whether economic, agricultural and climate-driven shocks lead to conflict (Liebig et al., 2022; Koubi, 2018; Forsyth & Schomerus, 2013). The risk category on the lack of infrastructural coping capacity provides an overview of communities' capacity to access public services, therefore indicating the infrastructural and economic viability of the localities they pertain (Marin-Ferrer et al., 2017). Low economic and infrastructural capacity to cope with shocks, and more broadly economic hardship, provide indications on poverty levels which are often ingrained with other factors interconnected with conflict (Liebig et al., 2022). Lastly, the human hazards risk category informs on the projected risk of conflict as well as its intensity, therefore evaluating the actual risk of conflict and the potential tensions there may be in given locations.

The tool draws on the growing body of research on environmental peacebuilding, which is the practice of using environmental challenges and resource-based disputes as opportunities to build cooperation, social integration, and peace through the transformation of natural resource management strategies (Krampe, Hegazi & VanDeveer, 2021). It employs environmental peacebuilding theories to express how climate adaptation can contribute to peacebuilding through a climate-peace framework (Sarzana et al., 2023). This framework lays the theoretical groundwork for connecting climate adaptation elements to peace co-benefits. The framework introduces six climate-peace mechanisms: economic development, building institutions, building trust and cooperation, resource sustainability, enhancing knowledge and building capacity and resilience. These climate-peace mechanisms are the ways through which climate adaptation can unify conflicting communities against shared insecurities. The climate-peace mechanisms translate how different characteristics of climate action interventions can mitigate conflict drivers, such as by strengthening livelihoods, improving resource governance, and addressing inequality and environmental degradation. Table 1 provides an overview of the climate-peace mechanisms and their components. Sub-mechanisms represent concrete activities and goals which can contribute to the main mechanism, and therefore to peacebuilding when integrated within climate adaptation interventions.

Table 1 Overview of climate-peace mechanisms, sub-mechanisms and examples of related climate action program activities

Climate-Peace Mechanisms	Climate-Peace Sub-Mechanism
Economic development	Create livelihoods and sustain existing ones
	Develop bi-communal spaces and infrastructures
	Foster the provision of public goods and services
Building institutions	Enhance institutional capacities for good environmental governance
	Facilitate legal pluralism and resource rights
	Foster equitable distribution of resources and benefits
Building trust and cooperation	Involve both high and grass-root levels while minimizing transboundary contacts in violent contexts
	Foster intercommunal trust and create shared identities
	Enhance social cohesion and empower vulnerable groups
Resource Sustainability	Restore degraded ecosystems
	Foster adoption of practices for sustainable use of resources
	Community-based conservation of ecosystems and common-pool resources
Enhancing knowledge	Raise public awareness and increase learning opportunities
	Establish the recognition of diverse ontologies in climate adaptation through grassroots approaches
Building capacity and resilience	Increase livelihood climate coping capacity
	Increase livelihood climate adaptation capacity
	Increase livelihood climate transformative capacity

Through literature reviews, these mechanisms were linked to JRC's crisis risk categories based on their relevance in addressing them (Sarzana et al., 2023). Risk categories are here also referred to as conflict and insecurity drivers. Table 2 shows the linkages between mechanisms and drivers stemming from this review indicating which mechanisms are critical for counteracting the drivers identified. The linkages are embedded within the logic of the CSST for featuring contextual programmatic priorities when designing climate adaptation interventions. For instance, the table shows that when designing a program, building institutions is a crucial component in addressing all drivers of conflict and insecurity, while resource sustainability appears to be crucial to address natural hazards, vulnerable groups and socioeconomic vulnerabilities.

Table 2 Linkages between climate-peace mechanisms and drivers of conflict and insecurity established by the Climate peace Framework. Tick marks show which climate-peace mechanisms are critical to tackle each conflict and insecurity driver

Drivers of conflict and insecurity	Climate-Peace Mechanisms					
	Economic development	Building institutions	Building trust and cooperation	Resource Sustainability	Enhancing knowledge	Building capacity and resilience
Low infrastructural capacities to cope with shocks	✓	✓				
Low institutional capacities to cope with shocks		✓	✓		✓	✓
Human hazards	✓	✓	✓			✓
Natural hazards	✓	✓		✓		✓
Vulnerable groups	✓	✓	✓	✓	✓	
Socio-economic vulnerabilities	✓	✓	✓	✓		✓

3. THE CLIMATE SMART VILLAGE APPROACH IN SÉGOU, MALI

3.1 Contextualizing Cinzana's municipality, Ségou region

The Cinzana municipality is situated in the central part of Mali, within the Segou region. It is located approximately 235 kilometers northeast of the national capital, Bamako. The municipality spans an area that includes a blend of savannah, woodland, and agricultural land. The Bani River, a major tributary of the Niger River, flows through the region, providing essential water resources for agriculture and daily life (World Bank, 2019). Cinzana experiences a classic Sahelian climate, characterized by a long dry season and a short, but intense, rainy season. Mali suffers from recurring severe droughts and floods and climate trends show that the average annual temperature has increased by 0.7 degrees since the 1960s, with a decline in rainfall since the 1950s, however precipitation levels have partially recovered (USAID, 2018). Rainfall in Cinzana is highly unpredictable and can vary significantly from year to year, leading to droughts, irregular rainfall patterns, and climate-induced stressors such as food and water scarcity, crop failures, and malnutrition (Ayité & Brown, 2009). There is a consensus that the negative impact of climate on agricultural production in Mali, and a Green Climate Fund concept note reported that this country is facing a drop in agricultural yields and a surge in food insecurity (Tchoupé Makougoum, 2018). Agriculture is the dominant economic activity in the region, with the majority of the population engaged in subsistence farming. Primary crops cultivated include millet, sorghum, maize, rice, and cotton. Livestock rearing, such as cattle and goats, is also a significant source of livelihood. However, limited access to modern farming techniques and climate-induced shocks frequently impact agricultural productivity. The combination of climate-induced agricultural challenges and a high dependency on rainfed agriculture has resulted in elevated food insecurity in the region. Irregular rainfall patterns and prolonged droughts lead to water scarcity for both crops and livestock, limiting the availability of essential food resources (UNDP Mali, 2020). Reduced crop yields and crop failures often result in reduced food availability at the household level, leading to malnutrition and food shortages (Ayité & Brown, 2009; World Bank, 2019). This not only affects food production but also exacerbates the vulnerability of the population, particularly smallholder farmers. The interplay between climate change, food insecurity, and poverty has created a vicious cycle, as impoverished communities have limited capacity to adapt to and mitigate the impacts of climate change on agriculture (Goudou & Sidibé, 2014). These issues contribute to increased migration from rural areas to urban centers in search of livelihoods, further straining the already limited resources and services available. While male migration is increasing in response to climate change, women are suffering immobility, which means they are responsible for historically male-dominated tasks such as livestock and forest maintenance, becoming more exposed to climate impacts and more vulnerable (Djoudi & Brockhaus, 2011; NRC, 2023; Tarif & Grand, 2021). In response to these challenges, communities in the Segou region, including Cinzana, have adopted various adaptation strategies. These include crop diversification to improve resilience, implementing water management practices, and developing early warning systems to mitigate climate-induced shocks (UNDP Mali, 2020). NGOs and government initiatives also play a role in supporting these adaptation efforts and building climate resilience in the agricultural sector (World Bank, 2019). In recent years, there has been an emphasis on diversifying livelihoods through non-agricultural sectors as well, such as handicrafts, small-scale trade, and artisanal gold mining (Goudou & Sidibé, 2014). Cinzana is home to diverse ethnic groups, including Bambara, Fulani, and Songhai populations, which sometimes leads to inter-ethnic tensions and conflicts over resources. Weak governance and limited access to basic services, including healthcare and education, present challenges for the local population. Mali's political instability, security concerns related to terrorism and insurgency, and broader governance and justice issues impact the region (Munro & Pendleton, 2019). Insecurity linked to jihadism-related violence has spread into Segou, which is also affected by cattle theft and banditry, and the proliferation of arms coming in from the north (Nagarajan et al., 2022). While the southern and western areas of Mali are comparably more stable, these regions are affected by land grabbing, and conflicts between farmers over land boundaries and farmer-pastoralist disputes leading to theft and violence (Nagarajan et al., 2022). Since 2020, violent conflicts in Mali have killed 2,070 people and displaced 300,000 people, forcing many to move to the south or towards neighboring countries (Targba, 2022; USAID 2019).

3.2 The Climate Smart Village approach in Cinzana

The Climate-Smart Village (CSV) approach is based on several Climate-Smart Agriculture (CSA) technologies and practices developed by the CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS). A CSV is a village in which farmers, researchers, and other partners test different CSA technologies and practices in an integrated and participatory manner to achieve climate adaptation, mitigation and food security (Tetteh et al., 2020). This program was implemented in Cinzana between 2012 and in 2021 and comprised of the following technologies and services (Bonilla-Findji et.al., 2017):

- CSA practices
    - Improved varieties of sorghum, millet, sesame and fonio (drought tolerant crop varieties)
    - Integrated Nutrient Management for sorghum and millet (Microdosing)
    - Organic manure
    - Intercropping of sorghum, millet and fodder crops (groundnut, cowpea)
    - Agroforestry intercropping: jatropha based agroforestry with sorghum and millet
    - Tree planting: Gliricidia sepium, Moringa olifera and Adansonia digitata
    - Water Harvesting through contour ridging of sorghum and millet
- Agro-meteorological services
    - Agroadvisories on fertilizer and pesticide application
    - Agroadvisories on varieties applied under the forecasted information
    - Seasonal forecast
- Financial services
    - Informal saving groups

This CSV program led to several positive outcomes, including farm productivity, food security, better livelihood conditions, improved soil fertility and positive gender outcomes such as women empowerment (Ouedraogo et al., 2018; Dembele and Magassa, 2018).

4. APPLICATION OF THE CSST

4.1 Step 1 - Conflict drivers' scores in Ségou, Mali

Step 1 of the tool defines the context by identifying the locally relevant potential drivers of conflict and insecurity (see appendix 1 for more detailed methodology). For this case study, all drivers of conflict and insecurity identified in step 1 scored high-level of risk. In this case study, scores are expressed at regional level, due the unavailability of more granular data within the JRC INFORM Risk database for the case of Mali. This implies that these scores do not appreciate the diversity in risk conditions across the different municipalities and communes within Ségou. According to these scores, societies in Ségou seem to be strongly affected by socioeconomic vulnerabilities, by the lack of both institutional and infrastructural capacities to cope with shocks, by natural and human hazards and to also present vulnerable groups (table 3).

Table 3 Scores for drivers of conflict and insecurity in Ségou (regional level scores)

Drivers	Score
Failing Infrastructures	0.67
Failing institutions	0.7
Human hazards	0.85
Natural hazards	0.68
Vulnerable groups	0.45
Socio-economic vulnerabilities	0.67

Information on the context of the Ségou region confirms these high scores. Mali has an abundance of human hazards, in the form of state-based, intrastate, and communal violence (UCDP, 2022; Kurath et al., 2022; Nagarajan et al., 2022). In the region of Ségou competition for resources between subsistence groups including farmers, herders and some fishermen is prevalent (CARE, 2023). Such competition translates into conflicts for resources over grazing, forest and arable land and theft of livestock; these land-based conflicts are magnified by a growing rate of deforestation and agricultural expansion (CARE, 2023). The presence of jihadist groups is a human hazard regardless of resource-related competition, and these armed groups grow by exploiting the grievances of pastoralists involved in farmer-herder disputes, further exacerbating these conflicts (CARE, 2023; Kurath et al., 2022). The high-risk level for institutional capacities to cope with shocks is explained by the weak governance in the region, featuring high corruption rates, political conflicts and poor natural resource management and conflict mitigation mechanisms in place (CARE, 2023; Munro & Pendleton, 2019). The natural hazards high-risk score captures the prevalent climate-related impacts in the region, especially on agriculture, including droughts, irregular rainfall patterns and an overall decline in precipitation (USAID, 2018). Information on the region also confirms the high-risk score for the infrastructural capacities to cope with shocks indicator, as this region is characterized by weak infrastructures, poor village electrification and landlocked communities (Ba and Bøås, 2017; Kurath et al., 2022; CARE, 2023). Socioeconomic vulnerabilities are also high, as poverty, unemployment and inequality rates are significant in this prevalently rural region, as in rural areas the percentage of people living in poverty amounts to 55.2 percent due to a lack of economic opportunities (INSTAT, 2017). Additionally, the high-scoring vulnerable groups indicator entails that malnutrition, food access limitations, health conditions and child mortality are salient, and that there are groups at a higher risk of being marginalized from financial and social services (Marin-Ferrer et al., 2017). This score can be confirmed as it is widely documented that this region is facing food insecurity challenges and great levels of pastoralist communities' social exclusion (UNDP Mali, 2020; Goudou & Sidibé, 2014; CARE, 2023; Nagarajan et al., 2022).

4.2 Step 2 - Climate-peace mechanisms for the proposed intervention

In Step 2, the proposed climate intervention program is scored according to its relevance to various climate-peace mechanisms, which are partitioned into sub-mechanisms. In this step, scores are given to each sub-mechanism indicating whether they are fulfilled (score of 1), somehow fulfilled (score of 0.5) or not fulfilled (score of 0). Overall climate-peace mechanisms scores are generated by averaging sub-mechanism scores (see appendix 1 for more detailed methodology). Table 4 displays the scores assigned to each sub-mechanism for the case of the CSV intervention in Cinzana with notes justifying these scores.

Table 4 Scoring system applied to the Climate Smart Village approach implemented in Cinzana

Climate Peace mechanisms	Sub-mechanisms	Indicators & examples/practices	Score	Notes
1. Economic development	Create livelihoods and sustain existing ones	**Secure food production:** provision of necessary inputs, irrigation sources, climate information **Diversify income and livelihood:** spread farm operations, mixed-systems approach, analyze market value chains to address bottlenecks and identify opportunities for added value **Restore degraded infrastructures**: sustain/introduce irrigation systems, mechanization technologies	1	Seasonal forecast, Improved Varieties, Intercropping, increased productivity (Bonilla-Findji et.al., 2017)
	Develop bi-communal spaces and infrastructures	**Introduce intercommunal infrastructures**: develop shared collecting/ storing/ processing/ transporting facilities for produce **Facilitate access to intercommunal resources**: extend fallow areas/pastures	1	Informal – saving groups (Bonilla-Findji et.al., 2017) Setting up of a storage warehouse has enabled the community to cope with lean seasons, especially in years of drought, resulting in an association
	Foster the provision of public goods and services	**Bolster equitable and efficient delivery of public services:** monitor funds allocation, increase availability of extension services **Increase government revenues from natural resource management:** increase available resources for the provision of public goods and services, foster foreign investment	0	No mention of public services deployed
2. Building institutions	Enhance institutional capacities for good environmental governance	**Address the illicit use of natural resources:** monitor protected areas/resources **Address the conflict economy:** reduce corruption, promote transparency **Build natural resource management governance, institutions, and capacities**: fortify subnational institutions, involve authorities in administration of program	0.5	Connecting PAR-CSA achievements to national policies through a science policy dialogue (Bayala et al., 2021) No mention of the fortification of local institutions
	Facilitate legal pluralism and resource rights	**Secure property rights:** map properties, address legal ambiguities on natural resource tenure and rights, certify resource rights **Deploy effective conflict management and resolution processes:** facilitate communication and negotiation around resources	0	No mention of property rights, legal ambiguities, and negotiation processes around resources implemented
	Foster equitable distribution of resources and benefits	**Regulate the use of and rights to resources more effectively and equitably:** make tenure governance policies more inclusive, transparent, and fair, strengthen the links between formal and informal natural resource management systems, reform natural resource management policies **Ensure program benefits are evenly distributed across groups:** all relevant actors concerned are made aware of the project and its benefits	0	No mention of governance policies and even distribution of program benefits No mention of the fortification of local institutions for good environmental governance
3. Building trust and cooperation	Involve both high and grass-root levels while minimizing	**Involve all community stakeholders in planning process:** meaningfully involve relevant community stakeholders (climate vulnerable and conflict-affected households) in the program's planning and administration processes, engage informal sector in	0.5	Building strong partnerships to co-design and develop agricultural systems that improve ecosystem and population resilience (Bayala et al. 2021)

	transboundary contacts	program development **Decrease opportunity cost for conflict:** address needs of both high and grass-root level figures in all groups targeted by the program, identify root causes of intercommunal conflict through participatory rural appraisal strategies **Participatory approach through minimized transboundary dialogue**: ensure participatory approaches do not gather program beneficiaries involved in active (violent) conflict together		No participatory approach mentioned.
	Foster intercommunal trust and create shared identities	**Address intercommunal power imbalances**: address disparities in wealth and negotiating capacity, support beneficiary communities through equity-based approaches rather than egalitarian ones **Create neutral spaces for dialogue:** gather program beneficiaries to analyze, collaborate, and find creative solutions for climate challenges, involve moderators in project plans	1	Gender power imbalances addressed: Kitchen gardening perimeters and training dedicated to women producers led to increased women earnings (Dembele and Magassa, 2018) Development of an association
	Enhance social cohesion and empower vulnerable groups	**Foster cooperative behavior:** enable communities to produce collective benefits, strengthen cooperatives, create opportunities for shared outputs, increase social capital in cooperative arrangements and networks **Involve on vulnerable groups in decision making processes:** include women, youth, marginalized, and disabled in program design and administration **Address gender-based discrimination:** include protocols to address gender-based violence, provide guidelines for reporting and confidentiality	0.5	Women incited to participate (Dembele and Dembélé, 2017) Key stakeholders involved in the project (researchers, farmers, development agents, and students) capacity strengthening through vocational and academic training (Bayala et al., 2021) Setting up of a storage warehouse has enabled the community to cope with lean seasons, especially in years of drought, resulting in the creation and development of an association enabling producers to save earnings and produce collective benefits
4. Resource sustainability				
	Restore degraded ecosystems	**Avoid future conflicts over scarce natural resources:** employ restoration frameworks for degraded landscapes and ecosystems, increase access to, availability and quality of water and land resources	1	Soil and water conservation techniques restoration of eroded fields: contour ridge tillage technique (Traore et al., 2017)
	Foster adoption of practices for sustainable use of resources	**Increase natural capital:** recycle resources, regenerative agricultural practices **Promote diversity in production systems:** increase land-use diversity or diversity at the landscape scale, Sustainable shifting cultivation, management of heterogeneous landscape, promote spatially diversified production systems through polycropping, promote rotational and regenerative grazing to improve soil quality and forage yield **Create bicommunal spaces for landscape restoration:** meaningfully involve communities in the planning of bi-communal NRM projects (dams, forest conservation…)	1	Integrated nutrient management, tree intercropping, intercropping (Bonilla-Findji et.al., 2017) Different sustainable land use practices mentioned
	Community-based conservation of ecosystems and common-pool resources	**Protect ecosystems and biodiversity:** identify local resources at risk (endangered/ vulnerable species), identify ecosystem services of valuable resources (cultural, provisioning, regulating, supporting services), plan conservation efforts for such resources **Establish conservation committees:** meaningfully involve the community in the definition and development of protected areas, and NRM projects more broadly, to address the needs of all resource users and prevent the exclusion of marginalized groups	0.5	Water harvesting, Integrated nutrient management, tree planting (Bonilla-Findji et.al., 2017) No mention of community-based common pool resource management
5. Enhancing knowledge				
	Raise public awareness and increase learning opportunities	**Provide technical knowledge on natural resource management:** tools, workshops and learning opportunities to help communities preserve dwindling or scarce resources **Provide learning opportunities and tools** to comprehend the risks posed by climate change, especially regarding slow onset processes	1	Agro- advisories on varieties applied under forecasted information Agro-advisories on fertilizer and pesticide application Providing information/awareness/advice to

		Increase public awareness to address violent conflict: environmental educational activities to develop sustainable and diversified livelihoods on environmental issues and environment-conflict linkages		farmers AND training of farmers (Traore et al., 2017) Gender power imbalances addressed: Kitchen gardening perimeters and training dedicated to women producers led to increased women earnings (Dembele and Magassa, 2018)
	Establish the recognition of diverse ontologies in climate adaptation through grassroots approaches	**Co-design programs to integrate local knowledges:** meaningfully involve representatives from local groups in the planning of the projects to include existing local climate adaptation and natural resource management strategies, ensure diverse knowledge holders have equal voice and that their strategic interests are addressed **Value traditional and indigenous knowledge:** seek to include traditional practices and customs in program development and recognize the legitimacy and value of indigenous and local knowledge **Provide knowledge sharing opportunities:** plan workshops and other community-engaging activities for different groups to share their traditional practices and beliefs in place to manage natural resources and adapt to climate changes	0.5	PAR-CSA principle to integrated participants in CSA definitions (Bayala et al., 2021) No mention of the integration of indigenous knowledge into the adaptation program
6. Building capacity and resilience				
	Increase livelihood climate coping capacity	**Identify local assets and needs:** understand impact of conflict and climate on livelihoods and production systems, provide social protection schemes to strengthen post-conflict human capital (health, nutrition, education, employment) **Decrease sensitivity of risk exposed areas:** include sensitivity analyses, identify maladaptive livelihood strategies based on sensitivity, provide early warning systems **Strengthen production systems coping capacity to shocks:** spread farm operations and diversify produce, adjust cropping/harvesting times, develop storage capacities, facilitate adoption of insurance-based schemes	0.5	Ridging tillage, crop diversification, storage capacities (Bonilla-Findji et.al., 2017) No activities for the prevention of flood-related and climate extreme events-related shocks No mention of activities to decreaes the sensitivity of risk exposed areas
	Increase livelihood climate adaptation capacity	**Increase adaptation capacity of social systems**: strengthen social capital (networks and connections) and financial capital to expand capacities to grow production systems (e.g.: financial services, credits), increase land-tenure security, address instrumental needs of communities	0.5	improving access to agricultural credit AND providing subsidies for inputs/equipment (Traore et al., 2017) No mention of support of social networks
		Increase adaptation capacity of production systems: facilitate the adoption of adapted crops/ cultivars & animal types/ breeds, improve crop residue management, integrated nutrient management, provision of post-harvest storage and water harvesting structures, mixed production systems, increase access to collection, refrigeration, processing and transportation infrastructures	1	Integrated nutrient management, Intercropping, ridging tillage (Bonilla-Findji et.al., 2017)
	Increase livelihood climate transformative capacity	**Address root causes of poverty and inequality:** strengthen sense of agency of vulnerable and marginalized groups, engage women and youth in long-term change processes that shift power, beliefs, values and ways of thinking and behaving to support greater levels of justice and equity **Equity, dignity, inclusion:** support fair, dignified and inclusive livelihoods for all actors engaged in food systems, especially small-scale food producers **Promote food sovereignty:** developing and informing policies and approaches that allow communities to decide the way food is produced, traded and consumed	1	women incited to participate, Gender power imbalances addressed: Kitchen gardening perimeters and training dedicated to women producers led to increased women earnings (Dembele and Magassa, 2018) Development of an association

4.3 Results

The results of the CSST are expressed visually through displaying ideal climate-peace mechanisms scores for the selected context next to the climate-peace mechanisms scores of the proposed intervention on spider charts (figure 1). Ideal scores highlight which climate-peace mechanisms should be prioritized in an intervention package, and how well matched the design of an intervention is to address conflict drivers. The ideal mechanism scores from step 1 are informed by contextual drivers of conflict and insecurity (table 3), and the scores of the intervention packages in step 2 derive from the scoring system of the climate adaptation intervention (table 4). Given the high levels of risk for most conflict and insecurity drivers identified in step 1, ideal climate-peace mechanisms scores for the region of Ségou mostly result high, all amounting to 80%. A gap between ideal and actual climate-peace mechanisms exists in Cinzana, with several ideal mechanisms being met by the mechanisms' scores delivered by the CSV program (figure 1). The CSV scored 83% for both resource sustainability and building trust and cooperation mechanisms, outscoring ideal conditions; the two mechanisms building capacity and resilience and enhancing knowledge scored 75%, somewhat meeting ideal conditions; the economic development mechanism only scored 67%; lastly, the building institutions one scored 17%, significantly underscoring ideal conditions.

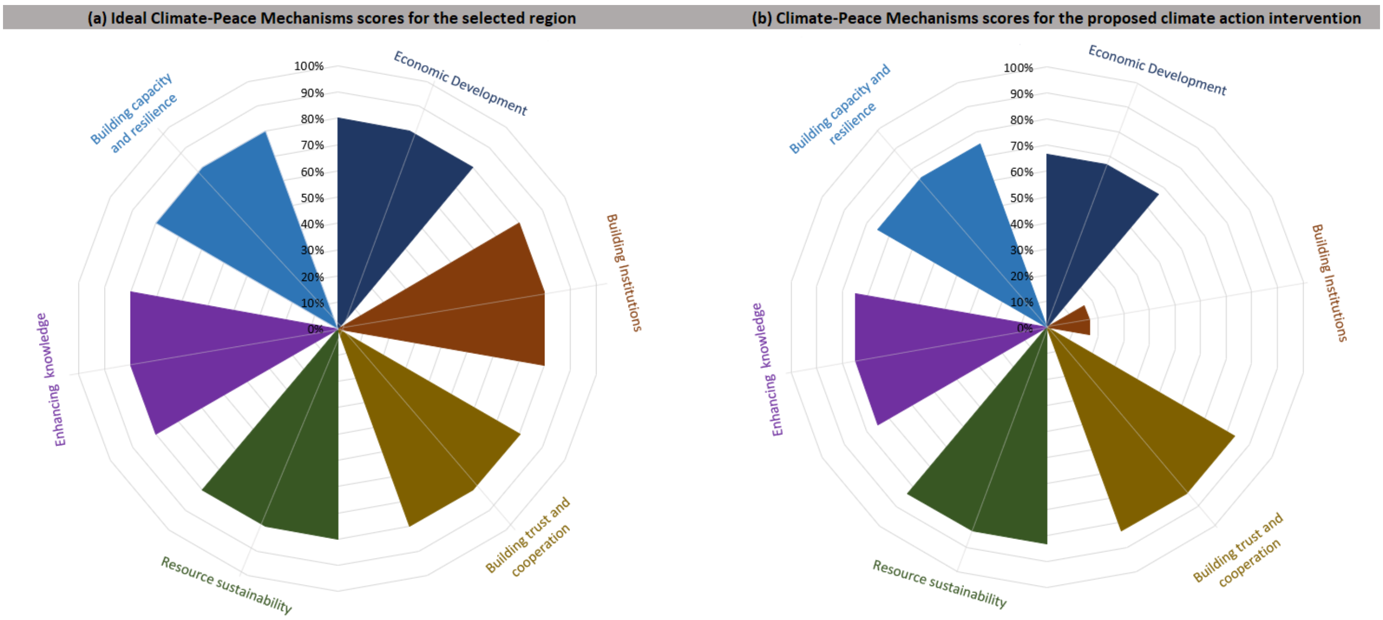

Figure 1 Spider charts displaying CSST results - ideal climate-peace mechanisms scores in Cinzana (left) and climate-peace mechanism scores for the CSV intervention (right)

4.4 Recommendations

Based on the results of the CSST arising from the comparison of ideal mechanisms' scores for the selected context (figure 1-a) and mechanisms' scores delivered by the proposed program (figure 1-b), recommendations can be drawn by targeting the mechanisms that do not meet ideal ones. Recommendations are formulated by advising the integration of the mechanisms' low-scoring components within the proposed adaptation intervention. From figure 1, it can be seen that the program should prioritize the integration of components to increase its building institutions mechanism score in particular, as well as its scores for economic development. The program could increase these scores by better integrating the sub-mechanisms that scored low (see red rectangles in figure 2).

In terms of institutional capacity building, the CSV intervention strives to promote transparency by connecting PAR-CSA achievements to national policies through a science policy dialogue (Bayala et al., 2021). However, it fails to fortify local institutions, secure property rights, address legal ambiguities, and to strengthen institutions for good environmental governance. Recommendations include mapping properties and addressing legal ambiguities on tenure rights, as well as integrating activities to foster an equitable distribution of resources and benefits and strengthening the links between formal and informal natural resource management systems. Also, reforming, strengthening, and enforcing natural resource management policies is crucial for the context of Ségou, where deforestation and agricultural expansion prevail and risk intensifying an illicit use of natural resources. These recommendations revolving around integrated institutional resource management could potentially prevent resource-related conflicts such as the ones between agricultural and pastoralist groups.

In terms of economic development, the CSV intervention creates livelihoods while sustaining existing ones and it also contribute to the development of bi-communal spaces and infrastructures. However, it fails to foster the provision of public goods and services. This could be achieved through further integrating programmatic components that strive to bolster equitable and efficient delivery of public services, such as by monitoring funds allocation and increasing the availability of extension services. In addition, regenerating resources would fuel their availability and usefulness in providing public goods and in increasing government revenues from natural resource management.

Mechanisms	Sub-mechanisms	Score	
Economic development	1.1. Create livelihoods and sustain existing ones	1	0.667
	1.2. Develop bi-communal spaces and infrastructures	1	
	1.3. Foster the provision of public goods and services	0	
Building institutions	2.1. Enhance institutional capacities for good environmental governance	0.5	0.166666667
	2.2. Facilitate legal pluralism and resource rights	0	
	2.3. Foster equitable distribution of resources and benefits	0	
Building trust and cooperation	3.1. Involve both high and grass-root levels while minimizing transboundary contacts	0.5	0.833333333
	3.2. Foster intercommunal trust and create shared identities	1	
	3.3. Enhance social cohesion and empower vulnerable groups	1	
Resource sustainability	4.1. Restore degraded ecosystems	1	0.833333333
	4.2. Foster adoption of practices for sustainable use of resources	1	
	4.3. Conserve ecosystems and common-pool resources	0.5	
Promotion of worldview and knowledge system plurality	5.1. Raise public awareness and increase learning opportunities	1	0.75
	5.2. Establish the recognition of diverse ontologies in climate adaptation through grassroots approaches	0.5	
Building capacity and resilience	6.1. Increase livelihood climate coping capacity	0.5	0.75
	6.2. Increase livelihood climate adaptation capacity	0.75	
	6.3. Increase livelihood climate transformative capacity	1	

Figure 2 Climate-Peace mechanisms and sub-mechanisms scores for the CSV intervention in Cinzana

5. CSST VALIDATION WORKSHOP

CGIAR Climate Security held a workshop in Ségou on the 14th of July 2023 to test the results of the CSST and reflect on the reliability of its recommendations. This workshop brought together different stakeholders based in Ségou and Cinzana to compare the recommendations provided by the CSST with their expert opinions and to reflect on the positive and negative implications that the Climate Smart Village (CSV) approach implemented in the village of Ngakoro may have had on security and peacebuilding. These stakeholders were 12 key informants on the different drivers of conflict and insecurity presented by the CSST and acquainted with the CSV project. These participants included representants from the Malian Institute of Rural Economy (IER), the regional Directorate for Animal Services, the regional Directorate for Water and Forest resources, local agriculture and meteorological services organizations, the Cinzana municipality, different development NGOs and universities (see appendix 2 for the full list of participants).

Picture 1. Workshop participants in Ségou

The CSST provides two main outputs: the prevalent drivers of insecurity, and recommendations based on the comparison between ideal climate-peace mechanisms scores for the selected context and the ones arising from scoring the proposed program. This workshop was divided into two exercises followed by discussion sessions: an exercise to validate the drivers' scores, and an exercise to validate the ideal climate-peace mechanisms. Lastly, participants were asked to discuss practices that would have improved the conflict sensitivity and peace responsiveness of the CSV program in Cinzana.

5.1 Defining reliable indicator scores of localized drivers of conflict and insecurity in Cinzana

The first exercise of the workshop consisted in mapping local drivers of conflict and insecurity for the context of Cinzana's municipality, and Ségou's district more broadly, to assess whether the indicators provided by JRC significantly portray risk conditions. Participants were subdivided into groups and asked to brainstorm on the risk factors comprising each driver, and to provide a risk score between 0 and 10 for each of these factors. To facilitate the scoring, participants were informed with low-, mid- and high-risk thresholds as well as with score examples from other countries. After having mapped risk factors and their scores for each driver, these scores were averaged out to estimate a risk score per driver. The scores appear on the key informants' row in table 5.

Picture 2. Workshop participants scoring drivers of conflict and insecurity in Cinzana

The risk levels defined during the workshop generally mirror the high-risk ones provided by JRC, except for the human hazard indicator to which workshop participants attributed a mid-level risk score. The risk levels relating to institutional capacity to cope with shocks (7), infrastructural capacities to cope with shocks (6.7) and socioeconomic vulnerability (6.54) mirror very precisely the ones provided by JRC. During the workshop, low institutional capacity to cope with shocks was characterized by risk factors including high levels of corruption, weak law enforcement, particularly regarding laws related to pastoral policies as part of the agricultural orientation law. Low infrastructural capacities to cope with shocks were characterized by poor road networks, inadequate health centers and schools, poor access to electricity and water, and insufficient communication infrastructures. Socioeconomic vulnerabilities listed by the participants included high poverty and low development indices levels, rural exodus, emigration, high aid dependency and high inequality rates.

The risk levels relating to human hazards (6.06) and natural hazards (5.88) feature a lower risk score than the ones featured by JRC, by a difference of 1.5 and 1 respectively. Human hazards where qualified by farmer-herder conflicts, communal conflicts, disregard of laws around natural resource management, land-based conflicts, including forest and arable land and conflicts between villages. Natural hazards were exemplified by droughts, increasing temperatures, violent winds, land degradation, pests and diseases outbreaks affecting crops and cattle, floods, decreasing biodiversity of fauna and flora, and soil erosion. When asked to comment on the difference between their human hazard score and JRC's, participants mentioned that JRC overestimated the score since, according to them, the only conflicts concerning the region are communal and resource-related ones and that in the region armed group-related violence is not present. A participant stated: "The exploitation of natural resources between villages is mostly what generates problems; one village is prohibited from cutting forest wood while another cuts; this is what causes conflicts between specific villages for the exploitation of natural resources." Another participant commented on the overestimation of the security context: "Today, we can see that there are red zones in the Ségou region, and external people assume that you can't live there. However, individuals who live there have a different perception, which is amplified by the media. When it comes to our human hazards, it all comes down to resource competition. This wasn't a problem in the past because rivers were numerous, and water lasted much of the year. Everyone wants to possess a pond now that they are scarce. There will be more community strife for that in the future." Regarding natural hazards, participants mentioned that the score is lower than the national average since the hazards only put pressure on crop production.

The risk levels relating to vulnerable groups (6.6) greatly overscore the one estimated by JRC, by 2.1 points. Vulnerable groups were characterized by high levels of food insecurity, high prevalence of groups affected by low sanitation conditions, unequal access to information and education, handicaps, an increasing prevalence of refugee communities and high levels of marginalization in general. For this driver of conflict and insecurity, participants stated that Cinzana is very much affected by factors of vulnerability and stressed that JRC scores do not portray these localized threats. A participant commented on this risk factor: "When it comes to the vulnerable groups score assessed by JRC, it's made with an outsider's eye lacking an appropriate appraisal. Those of us who live here feel what we endure. During the drought of 2021, for example, there was no food, the harvest was catastrophic, and grain prices soared (millet, 500FCFA /kg). Not everyone had food, and in general everyone felt the drought. Also, Ségou is in the middle between the Southern part of Mali and the Northern, meaning it's the first refuge zone for people escaping violence inflicted by armed groups before heading South. We host a lot of uprooted people and refugees."

Table 5 Drivers of conflict and insecurity scores as provided by JRC and by the workshop participants (very low and low scores in green, mid-risk scores in yellow, high and very high scores in red)

	Low infrastructural capacities	**Low institutional capacities**	**Human hazards**	**Natural hazards**	**Vulnerable groups**	**Socio-economic vulnerabilities**
JRC INFORM scores	6.7	7	8.5	6.8	4.5	6.7
Key informants scores	7	7	6.06	5.88	6.6	6.54

5.2 Defining ideal climate-peace mechanism scores for the context of Cinzana and Ségou

The second exercise of the workshop consisted in mapping the relationships between climate-peace mechanisms and drivers of conflict and insecurities. This exercise aimed at validating the connections in table 2 arising from a literature review assessing which climate-peace mechanisms presented by the climate-peace framework (Sarzana et al., 2023) are crucial for addressing the drivers of conflict and insecurity presented by JRC as risk factors (Marin-Ferrer et al., 2017). To do so, participants were presented with a poster featuring drivers on one side, and mechanisms on the other, and were asked to link them based on their programmatic experience and knowledge on field, using references and examples in relation to the CSV implemented in Cinzana. These linkages assessed whether mechanisms are crucial, somehow useful, or not necessary for addressing those drivers. Straight lines were drawn when the relationship between a mechanism and a driver was evident and crucial, a dotted line when the relationship was present, but the mechanism did not appear to the participants as evident as for straight lines, and no lines when the relationship was absent. Table 6 summarizes the results of this exercise, showing which connections they found to be crucial (1), somehow important (0.5) and not important (0).

Table 6 Relationships between climate-peace mechanisms and drivers of conflict and insecurity resulting from the workshop

Climate Peace Mechanisms / Drivers of conflict	Economic Development	Building Institutions	Building trust and cooperation	Resource sustainability	Enhancing knowledge	Building capacity and resilience
Low infrastructures	0.8333	0.6667	0.5	0.6667	0.5	0.5
Low institutions	0.6667	1	0.6667	0.5	0.8333	0.5
Human hazards	1	0.6667	0.6667	1	0.6667	0.8333
Natural hazards	0.6667	0.6667	0.5	0.8333	0.8333	0.8333
Vulnerable groups	1	0.6667	1	1	0.8333	0.8333
Socioeconomic vulnerabilities	1	0.6667	1	1	0.8333	0.8333

During this exercise, economic development was defined as the most critical mechanism by the participants as it received the higher score in terms of importance for addressing all drivers of conflict and insecurity, followed by resource sustainability, enhancing knowledge, while the mechanisms building capacity and resilience, building institutions and building trust and cooperation were positioned on the same level. Compared to the linkages made in the climate-peace framework (table 2), the participants similarly prioritized economic development for addressing most drivers. The main difference between the framework's connections (table 2) and the ones made by the workshop participants is that fewer and weaker connections linked the enhancing knowledge and the resource sustainability mechanisms to drivers of conflict and insecurity in the framework. Participants gave much more importance to these mechanisms, meaning that according to them, raising public awareness, providing learning opportunities and natural resource conservation and sustainability were crucial components to address drivers of conflict in the regions. Additionally, the workshop participants attributed less importance to building institutions as a factor addressing drivers of conflict than in the framework. For the remaining mechanisms, the linkages with drivers did not vary significantly.

5.3 Workshop results

At the end of the workshop, participants were shown the ideal climate-peace mechanisms' scores arising from the insights given throughout the two exercises and were invited to comment on the differences with the ones presented by the CSST for the context of Ségou. Figure 1-a features the ideal mechanisms arising from the CSST framework, while figure 3-a features the ideal mechanisms arising from the participants' inputs throughout the exercises. Ideal climate-peace mechanisms scores in figure 3 are highly coherent to the ones in figure 1, revealing that peace co-benefits and conflict-prevention activities in climate adaptation programs are highly needed in the context of Cinzana as much as in the Ségou region. In figure 3 it can be seen that all climate-peace mechanisms are nearly as equal in importance, with ideal scores ranging from 80% to 81.5%. Compared to the ideal scores given by the CSST, the ideal mechanisms that slightly scored higher is the building trust and cooperation one, scoring 1% higher. When asked to reflect and comment on the ideal mechanisms' scores participants stated that enhancing knowledge and building capacity and resilience should be given more importance knowing the challenges of the different livelihoods in Cinzana.

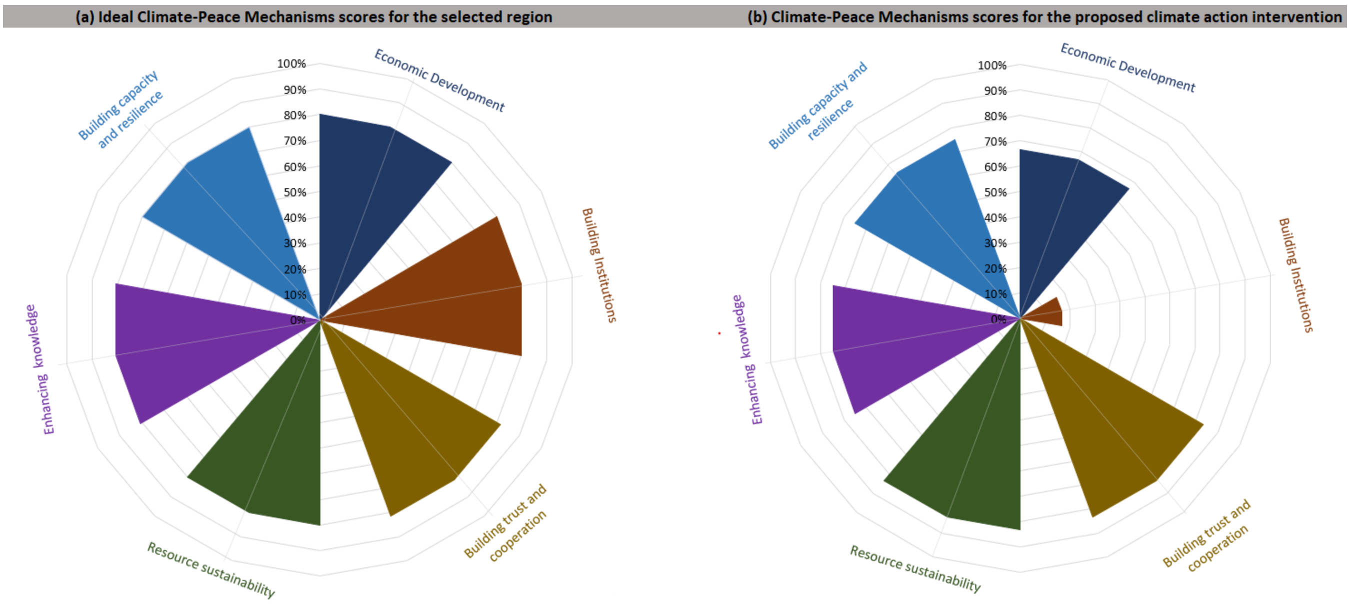

Figure 3 Spider charts displaying the CSST results with inputs from the workshop participants - ideal climate-peace mechanisms scores in Cinzana (left) and climate-peace mechanism scores for the CSV intervention (right)

By changing the ideal mechanisms scores, the recommendations formulated by the CSST in subsection 4.4 should also change. However, since the ideal mechanisms provided by the CSST do not differ from the ones developed through the 2 workshop exercises, recommendations remain the same: substantially integrating programmatic components for building institutions, and further incorporating economic development aspects. Based on these results, the CSV program should have better focused on targeting environmental governance objectives in the context of Cinzana, as well as some economic development ones. In particular the CSV should have better focused on strengthening the enforcing capacity of local institutions, particularly regarding natural resource protection, securing property rights and addressing legal ambiguities between neighboring communities. In terms of economic development, recommendations include further integrating programmatic components that strive to bolster equitable and efficient delivery of public services, such as by monitoring funds allocation and increasing the availability of extension services.

At the end of the workshop, participants were asked what the CSV program has achieved and what it should have been focusing on more to contribute to peacebuilding outcomes. In terms of achievements, participants mentioned the increased delivery of inputs, including fertilizers and improved seeds, of climate smart agricultural technologies, of agricultural extension agents and staff capacities, as well as the improved communication of climate and weather advisories through issuing mobile phones to farmers. These achievements have positively improved production and income of farmers and increased food security and climate resilience amongst households in Cinzana. These achievements alone can be considered in line with conflict sensitivity, as they reduce poverty and vulnerability, and prevent future conflicts over resources by securing productivity. Climate action and its contributions to reduced poverty address basic human needs, and thus creating livelihoods and sustaining existing ones, while increased farmers capabilities is a key element for peacebuilding and human security, since it represents options and instruments to face, mitigate and adapt to threats posed to human, environmental and social rights. In addition, participants mentioned that the CSV program addressed gender inequality structures present through the establishment of market gardening perimeters. These perimeters have enabled women to sell their market garden produce, earn more money and then buy cattle for labor, which has eased their work and transport tasks and improved their purchasing power. Also, the setting up of a storage warehouse has enabled the community to cope with lean seasons, especially in years of drought, and has encouraged producers to collaborate more in the management of foodstuffs. Such enhanced collaboration has resulted in the development of an association and ultimately better social cohesion. The inclusion of the above-listed programmatic components led to peace contributing and conflict preventing outcomes.

When asked about the features that should have been further incorporated within the CSV, participants mentioned the weak focus that the program attributed to improving natural resource management from an institutional point of view. They added an example: "The lack of programmatic activities targeting natural resource governance has not prevented unwanted consequences such as deforestation. Trails to disenclave the locality had been laid simultaneously by another project, and as a result, illegal extraction of natural resources increased as trucks were able to circulate more easily in the village and stock up on wood. Deforestation in this case resulted in mild conflicts over timber resources and non-timber forest products. In this specific case, if the CSV had allocated more importance to local governance capacities, these effects and conflicts over resources could have been potentially mitigated." Furthermore, the lack of awareness-raising initiatives in the CSV decreased the project's longevity, as recipients still lacked resource conservation capacities at the project's conclusion. Participants emphasized that people in Cinzana still require capacity building to appropriately manage the resources on which they rely, and that inadequate resource management will lead to a rise in conflicts.

The programmatic features listed by the participants resonate with the recommendations provided by the CSST in figure 2, resonating with the need to prioritize institution building features in the program as a way to prevent resource-related conflicts in a context characterized by dwindling forest and water resources.

6. CONCLUSION AND LIMITATIONS

This report, based on the case study in Cinzana, Mali, serves a dual purpose: to validate the Climate Security Sensitivity Tool (CSST) and to demonstrate its adaptability to localized contexts. The participatory workshop conducted in Ségou in July 2023 unveiled important insights and affirmed the value of employing the CSST in real-world applications.

In the case study of Cinzana, a municipality in the Ségou Region of Mali, the CSST provided valuable insights into the region's specific challenges and vulnerabilities. A context analysis through the tool’s first step revealed a complex landscape characterized by low infrastructural and institutional capacities, and various natural and human hazards, as well as ingrained socioeconomic vulnerabilities and vulnerable groups. These challenges are further exacerbated by insecurity related to armed groups present across the country, by factors such as poverty, unemployment, inequalities, and climate stressors like increased temperature, decreased precipitation levels, erratic rainfall patterns, and seasonal variability.

The Climate-Smart Village (CSV) approach is the adaptation intervention that was tested with the CSST. This program, which ran from 2012 to 2021, aimed to enhance climate adaptation, mitigate climate change, and improve food security. The CSV intervention achieved notable success in increasing farm productivity, enhancing food security, reducing male out-migration, and boosting education rates.

The CSST validation workshop involved local stakeholders and experts, who critically assessed the CSST's results and recommendations. The workshop validated the drivers of conflict and insecurity, emphasizing that the risk factors in the Ségou must better portray the wide prevalence of vulnerable groups and pointed to the fact that human hazards are more related to communal and resource access disputes rather than terrorism in Ségou. The exercises carried throughout the workshop revealed a strong alignment between the ideal climate-peace mechanisms outlined by the CSST and those developed by the participants, indicating a parallel importance of peace co-benefits and conflict-prevention activities within climate adaptation programs in the Ségou region and more specifically in Cinzana. Both the CSST results and the workshop participants suggested that certain aspects of the program could have been further enhanced to increase its conflict sensitivity and peace responsiveness. The lack of emphasis on institutional natural resource management and insufficient awareness-raising initiatives are cited as areas where the program fell short. This oversight resulted in challenges like deforestation, increased illegal extraction of resources, and potential conflicts over natural resources, emphasizing the need for stronger governance capacities and sustained educational efforts.

Importantly, these participant-suggested programmatic features resonate with the recommendations of the CSST, emphasizing the significance of prioritizing institution-building elements within programs to prevent resource-related conflicts, especially in contexts facing environmental resource depletion.

Thereafter, the outcomes of the workshop reaffirmed that the CSST, when enriched with inputs from local stakeholders and experts, can provide recommendations tailored to address the unique challenges and vulnerabilities within a specific context. This flexibility and responsiveness to local insights underscore the tool's practicality and relevance in the field.

In conclusion, the workshop's outcomes validated the CSST's utility and underscored its adaptability to local contexts. It demonstrated that the tool can provide suitable recommendations when coupled with localized insights, enhancing its value in crafting context-specific climate adaptation programs that effectively address the drivers of conflict and insecurity in vulnerable regions.

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8. APPENDICES

Appendix 1. CSST step-by-step guide

The CSST is composed of two main steps: the context definition and the climate action scoring system. Implementing the first component results in the projection of the ideal set of climate-peace mechanisms for the selected context, while the second component provides the set of mechanisms currently delivered by the proposed program design. Visually aligning these two sets allows practitioners to re-define their intervention to match the ideal mechanisms.

The context definition step enables to identify the geographical location where a climate adaptation intervention is expected to be implemented. In this step, the user provides location-specific information (country – region – municipality), which enables risk indicator scores to appear. Indicators are sourced from the Joint Research Centre's INFORM risk indexes and are assessed on a scale from 0 and 10, with risk-level thresholds varying between categories (Marin-Ferrer et al., 2017). These risk threshold classes provide a weight of severeness of the drivers of conflict and insecurity. Drivers featuring very high risk are assigned a weight of 5, high risk is assigned a weight of 4, medium risk is assigned a weight of 3, low risk is assigned a weight of 2, and very low risk is assigned a weight of 1. Through the Climate Peace Framework, these drivers have been linked to climate-peace mechanisms based on their relevance in addressing them. Risk severeness coefficients affect the score of the mechanisms they have been linked to: when a driver features high risk, the mechanisms that are linked to it get a higher relevance score, while when the driver features low risk, its related mechanisms become less relevant. Relevance scores for each climate-peace mechanism are projected on a spider chart on a percentage scale informing users of the ideal scores for the selected context. This chart allows practitioners to identify the programmatic areas that require more attention – higher percentage scores – in order to prevent conflicts and contribute to peacebuilding.

The climate action scoring system step enables to define the extent to which the proposed adaptation program design performs for each climate-peace mechanism. In this step, the user scores the proposed climate-adaptation intervention across the sub-mechanisms within each climate-peace mechanism. Examples and proxies are provided for the user to get an understanding of the characteristics a climate-adaptation intervention needs to include in order to fulfill those sub-mechanisms. The user can only fill the sub-mechanisms scores with either a 1, a 0.5, or a 0: a score of 1 can be added when the sub-mechanism is fully fulfilled, a score of 0.5 when the sub-mechanisms is partly or indirectly fulfilled, and a score of 0 when the sub-mechanism is not fulfilled. These scores are then averaged out for each mechanism and projected on a spider chart on a percentage scale. The percentage scores inform users of the performance of their program for each component.

Based on the percentage results of the two spider charts, the user can detect which components in their program fall short by comparing ideal climate-peace mechanism scores to the ones delivered by the proposed program design. If a mechanism scores very high in the ideal conditions chart, it should also score high in the chart illustrating mechanisms scores delivered by the program design. When this is not the case, the user can further integrate programmatic features to increase the underscoring mechanism. For further information, Sarzana et al. (2023) provides a more detailed description of the CSST methodology.

Appendix 2. List of participants

No.	NAME OF PARTICIPANT	ORGANIZATION
1	Souaïbou Touré	Mairie Cinzana
2	Sababou Cisse	Conseil régional de Ségou
3	Seydou Coulibaly	TOGO TILE Ségou
4	Bréhima Bangaly	ONG Geres
5	Zoumana Kané	Université de Ségou
6	Fousseyni Djiré	Mali Météo
7	Fousseyni Diabate	Direction Régionale des Eaux et Forêts
8	Sidy Mody Sidibe	Direction Régionales des Services Vétérinaires Ségou
9	Sana Kassogué	Service local d'Agriculture
10	Dr. Siaka Dembélé	IER Cinzana
11	Daou Rokia Coulibaly	Direction Régionale de l'Agriculture Ségou
12	Marcel Daou	Personne ressource

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